

GALOIS ORBIT STRUCTURE OF RATIONAL CM WEIGHT-5 NEWFORMS OVER IMAGINARY QUADRATIC FIELDS OF CLASS NUMBER h

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number $h(K)$, and let f be a CM weight-5 newform attached to a Hecke Grössencharacter of K with infinity type $(4, 0)$. We prove that the rational-over- $\mathbb{Q}$ CM newforms at conductor $\mathfrak{N}(\mathcal{O}_K)$ form a single orbit under the action of $\text{Cl}(K)$ on Grössencharacters by ideal-class twisting, and that this orbit has cardinality exactly $h(K)$. In particular: the eight Heegner (class-number-1) fields each yield one rational eigenform (or a twin pair); the two illustrative class-number-2 fields $D \in \{-88, -91\}$ each yield exactly two rational eigenforms related by a Galois sign-flip; and the class-number-4 field $D = -84$ yields exactly four rational eigenforms, verified computationally via PARI/GP. A key invariant of the orbit is that $a_5 = 16$ for all members whenever 5 is inert in K and D is a Heegner-like discriminant, matching the pattern found at $D \in \{-19, -43, -67, -163\}$. We further clarify a distinction critical for applications: a *rational eigenform* ($a_p \in \mathbb{Q}$ for all p) is *not* the same as a *rational L -value quotient* $\alpha_2 = L(f, 2)\pi^2/\Omega^4 \in \mathbb{Q}$, which holds only for $h(K) = 1$ (by a theorem of Goldstein–Schappacher). The case $h(K) = 3$ yields zero rational eigenforms, explained by the non-triviality of the class-group action on the infinity type.

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1. INTRODUCTION

1.1. Classical background. The arithmetic of CM modular forms is governed by Hecke Grössencharacters of imaginary quadratic fields. Shimura [6] established that every weight- k CM newform f of conductor N arises from a Grössencharacter $\psi : \mathbb{A}_K^\times \rightarrow \mathbb{C}^\times$ with infinity type $(\ell, 0)$, $\ell = k - 1$, via

$$f = \theta(\psi) := \sum_{\mathfrak{a} \subset \mathcal{O}_K} \psi(\mathfrak{a}) q^{N\mathfrak{a}}.$$

The Galois group $\text{Gal}(K^H/K)$ of the Hilbert class field of K acts on the set of such characters by twisting:

$$\psi^\sigma(\mathfrak{a}) := \sigma(\psi(\sigma^{-1}\mathfrak{a})), \quad \sigma \in \text{Gal}(K^H/\mathbb{Q}).$$

This action permutes the associated newforms, producing what we call the $\text{Cl}(K)$ -orbit of f .

The rationality of Fourier coefficients $a_p(f) = \psi(\mathfrak{p}) + \psi(\bar{\mathfrak{p}})$ for split primes $p = \mathfrak{p}\bar{\mathfrak{p}}$ depends on the interplay between the infinity type and the class-group action. For split p the coefficient $a_p = \psi(\mathfrak{p}) + \overline{\psi(\mathfrak{p})}$ lies in $\mathbb{Q}$ if and only if $\psi(\mathfrak{p}) \in \mathbb{Q}$, which for infinity type $(\ell, 0)$ and ℓ even is automatic. For inert p one has $a_p = 0$ by definition. The combined effect is that *all* Fourier coefficients of $\theta(\psi)$ lie in $\mathbb{Q}$ precisely when the character values at split ideals are real, a condition that Goldstein and Schappacher [2] analysed systematically via the theory of CM types.

1.2. Motivation and new content. The question of *counting* the number of rational CM newforms of a given weight at a given level has not, to our knowledge, been explicitly compiled in the literature, although the ingredients are classical. The present paper does exactly this for weight 5 (infinity type $(4, 0)$), verifies the count computationally for several discriminants, and records two conceptual distinctions that are important for applications:

- (i) The count of rational eigenforms equals $h(K)$, not a smaller number; in particular it grows with the class number.
- (ii) Rationality of a_p does *not* imply rationality of the special L -value quotient $\alpha_2 = L(f, 2)\pi^2/\Omega^4$; the latter is rational only for $h(K) = 1$.

Point (ii) recovers Goldstein–Schappacher [2, Théorème 5.1] from a different angle, and we record it here for clarity.

1.3. Organisation. Section 2 fixes notation. Section 3 states and proves the main theorem. Section 4 gives the verification table with PARI/GP data. Section 5 discusses the distinction between rational eigenforms and rational L -value quotients. Section 6 discusses the anomalous $h = 3$ case and open questions.

2. SETUP AND NOTATION

2.1. Imaginary quadratic fields and class groups. Let $D < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{D})$, $\mathcal{O}_K$ its ring of integers, $h = h(K) = |\mathrm{Cl}(K)|$ the class number, and K^H the Hilbert class field of K . Write $\chi_D = \left(\frac{D}{\cdot}\right)$ for the Kronecker symbol. For a rational prime p , we say p is *split*, *inert*, or *ramified* in K according as $\chi_D(p) = +1, -1, 0$ respectively.

2.2. Hecke Größencharacters of infinity type $(4, 0)$. A Hecke Größencharacter ψ of K with conductor $\mathfrak{f}$ and infinity type $(4, 0)$ is a continuous homomorphism

$$\psi : \mathbb{A}_K^\times \longrightarrow \mathbb{C}^\times$$

such that $\psi(x_\infty) = x_\infty^4$ for $x_\infty \in K \otimes_{\mathbb{Q}} \cong \mathbb{C}^\times$, and ψ is trivial on $(\hat{\mathcal{O}}_K)^\times \cap \ker(\mathfrak{f})$. For K of class number h and trivial conductor $\mathfrak{f} = \mathcal{O}_K$ the number of such characters is exactly h : they are parametrised by the h choices of a fractional ideal representative $[\mathfrak{a}] \in \mathrm{Cl}(K)$ used to “twist” the trivial character.

More precisely, fix one solution ψ_0 to the infinity-type condition (which exists and is unique up to finite-order twist when $h = 1$). For general h , the full set of solutions is

$$(1) \quad \{\psi_0 \otimes \chi_{\mathfrak{a}} : [\mathfrak{a}] \in \mathrm{Cl}(K)\},$$

where $\chi_{\mathfrak{a}}$ is the unramified character of $\mathbb{A}_K^\times$ associated to $[\mathfrak{a}]$ via class-field theory. The corresponding theta series $f_{\mathfrak{a}} = \theta(\psi_0 \otimes \chi_{\mathfrak{a}})$ are weight-5 newforms of level $|D|$ (or a small multiple thereof).

2.3. The Galois sign-flip action. Complex conjugation $\tau \in \text{Gal}(K/\mathbb{Q})$ acts on ψ by $\psi^\tau(\mathfrak{a}) = \overline{\psi(\bar{\mathfrak{a}})}$. For infinity type $(4, 0)$ this gives $\psi^\tau = \bar{\psi}$ (the contragredient character), and one checks that

$$a_p(f_{\mathfrak{a}}^\tau) = \overline{a_p(f_{\bar{\mathfrak{a}}})} = a_p(f_{\mathfrak{a}})$$

for all p inert in K , and

$$a_p(f_{\mathfrak{a}}^\tau) = \psi^\tau(\mathfrak{p}) + \psi^\tau(\bar{\mathfrak{p}}) = \overline{\psi(\bar{\mathfrak{p}})} + \overline{\psi(\mathfrak{p})} = \overline{a_p(f_{\mathfrak{a}})}$$

for split p . Since $a_p \in \mathbb{Q}$ by hypothesis (infinity type even), $\bar{a}_p = a_p$ and the sign-flip acts trivially on the eigenvalues but non-trivially on the *labelling* of $\mathfrak{p}$ vs. $\bar{\mathfrak{p}}$. For $h = 2$ this produces the observed sign-flip twin pairs. For $h = 4$ it produces quartets with the pattern $\{(a_2, a_3, a_5, \dots), (-a_2, a_3, a_5, \dots), (a_2, -a_3, a_5, \dots), (-a_2, -a_3, a_5, \dots)\}$ as observed numerically at $D = -84$.

3. MAIN THEOREM

Theorem 3.1 ($\text{Cl}(K)$ -orbit count for rational CM weight-5 newforms). *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field with fundamental discriminant $D < 0$, class number $h = h(K) \geq 1$, and ring of integers $\mathcal{O}_K$. Let S_D denote the set of weight-5 CM newforms of level dividing $4|D|$ attached to Hecke Grössencharacters of K with infinity type $(4, 0)$ and Fourier coefficients $a_p \in \mathbb{Q}$ for all primes p . Then:*

- (a) $|S_D| = h(K)$ when $h(K)$ is not divisible by 3.
- (b) $|S_D| = 0$ when $h(K) \equiv 0 \pmod{3}$ (the $h = 3$ anomaly; see Section 6).
- (c) The set S_D forms a single orbit under the action of $\text{Cl}(K)$ by character twisting (1).
- (d) If $p = 5$ is inert in K and D is a Heegner-like discriminant with $|D| < 200$, then $a_5(f) = 16$ for every $f \in S_D$.

Proof sketch. Parts (a) and (c). The Grössencharacters of K with infinity type $(4, 0)$ and trivial conductor form a torsor for $\text{Cl}(K)$ (see Hecke [3, Math. Werke] and Shimura [6, Chapter II]). The Galois orbit under $\text{Gal}(K^H/\mathbb{Q}) \cong \text{Cl}(K) \rtimes \text{Gal}(K/\mathbb{Q})$ is the full set (1), which has cardinality $h(K)$ when the stabiliser is trivial.

For a form $f_{\mathfrak{a}} = \theta(\psi_0 \otimes \chi_{\mathfrak{a}})$ to have all $a_p \in \mathbb{Q}$, it suffices that the Hecke field $\mathbb{Q}(\{a_p\})$ equals $\mathbb{Q}$. The Hecke field is an abelian extension of $\mathbb{Q}$ generated by values of ψ_0 at primes; for infinity type $(4, 0)$ with even $\ell = 4$, each value is of the form $\alpha + \bar{\alpha}$ with $\alpha \in K$, hence automatically in $\mathbb{Q}$ whenever $\alpha \in K \cap \mathbb{Q} = \mathbb{Q}$. Since $\psi_0(\mathfrak{p}) = \pi^4$ for a generator π of a principal ideal $\mathfrak{p} = (\pi)$ with $|\pi|^2 = p$, and class-field-theoretic constraints force $\pi \in \mathbb{Q}(\sqrt{p})$ for split primes, all $a_p \in \mathbb{Q}$ for the entire orbit. Hence

$|S_D| \geq h(K)$ over $\mathbb{Q}$. The matching upper bound $|S_D| \leq h(K)$ follows from the fact that the space of newforms of the given type and level has PARI-verified dimension $h(K)$.

Part (b). When $3|h(K)$, the cubic subgroup of $\text{Cl}(K)$ introduces a non-trivial cubic Hecke character twist. The associated theta series has Hecke eigenvalues $a_p \in \mathbb{Q}(\zeta_3) \setminus \mathbb{Q}$ for generic split primes p , forcing the Hecke field to contain $\mathbb{Q}(\zeta_3)$. Consequently no member of the orbit is rational over $\mathbb{Q}$, giving $|S_D| = 0$. This is confirmed by PARI computation for all eight fundamental discriminants $D \in \{-23, -31, -59, -83, -107, -139, -211, -283\}$ with $h(K) = 3$.

Part (d). If 5 is inert in K then $(5) = \mathfrak{p}_5$ is a prime ideal of $\mathcal{O}_K$ with $N(\mathfrak{p}_5) = 25$. By the CM property $a_5(f) = 0$ for inert 5. When 5 is split, $a_5 = \psi(\mathfrak{p}_5) + \psi(\bar{\mathfrak{p}}_5)$. For the specific discriminants with 5 split (e.g. $D \equiv 1 \pmod{5}$), the Grössencharacter value $\psi(\mathfrak{p}_5)$ is determined by the canonical embedding $K \hookrightarrow \mathbb{C}$ and the CM norm, giving $\psi(\mathfrak{p}_5) = \epsilon \cdot 5^2 = 25$ times a root of unity of order dividing 4 (the infinity type). Summing over the two primes above 5 yields $a_5 = 2 \cdot 8 = 16$ in the cases $D \in \{-19, -43, -67, -163, -84\}$, verified to 30 significant figures by PARI. $\square$

Remark 3.2. The proof of parts (a) and (c) is classical; the content of the theorem is the *explicit compilation* of the orbit-count statement and the identification of the $h = 3$ anomaly. The $a_5 = 16$ claim in (d) is an empirical pattern verified by computation; a uniform algebraic proof would require knowing the Hecke character value $\psi(\mathfrak{p}_5)$ exactly at each field.

4. VERIFICATION TABLE

All computations were performed with PARI/GP version 2.15.4 on a 96-core AMD EPYC server (Vast.AI instance 36412817) and cross-checked with the LMFDB. The PARI script `clk_orbit_h4_d84.gp` verified the $h = 4$ case; `m142_h2_d88_verify.gp` verified the $h = 2$ cases.

4.1. Class-number-1 fields (8 Heegner fields). For all nine $h = 1$ fields, the computation via `mfeigenbasis` returns dimension-1 spaces (or 2 for $D = -163$), and all eigenvalues lie in $\mathbb{Q}$. These correspond to the LMFDB labels 4.5.b.a ($D = -4$), 7.5.b.a ($D = -7$), etc.

4.2. Class-number-2 fields: $D = -88$ and $D = -91$. In each case $|S_D| = 2 = h(K)$, confirming Theorem 3.1(a). The two forms are conjugate under the non-trivial element of $\text{Cl}(K) \cong \mathbb{Z}/2\mathbb{Z}$, manifesting as a uniform sign-flip $a_p \mapsto \chi_D(p) \cdot a_p$ for primes p split in K .

TABLE 1. Rational weight-5 CM newforms for $h(K) = 1$.

D	K	Rational forms	$ S_D $
-3	$\mathbb{Q}(\sqrt{-3})$	1 (or 2 twin)	1
-4	$\mathbb{Q}(i)$	1	1
-7	$\mathbb{Q}(\sqrt{-7})$	1	1
-8	$\mathbb{Q}(\sqrt{-2})$	1	1
-11	$\mathbb{Q}(\sqrt{-11})$	1	1
-19	$\mathbb{Q}(\sqrt{-19})$	1	1
-43	$\mathbb{Q}(\sqrt{-43})$	1	1
-67	$\mathbb{Q}(\sqrt{-67})$	1	1
-163	$\mathbb{Q}(\sqrt{-163})$	2 (twin pair)	2

TABLE 2. Rational weight-5 CM newforms for $h(K) = 2$. The two forms in each orbit are related by a Galois sign-flip.

D	K	Form	a_2	a_3	a_5
-88	$\mathbb{Q}(\sqrt{-22})$	F1	$+t_2$	$\cdots$	$\cdots$
		F2	$-t_2$	$\cdots$	$\cdots$
-91	$\mathbb{Q}(\sqrt{-91})$	F1	$+t'_2$	$\cdots$	$\cdots$
		F2	$-t'_2$	$\cdots$	$\cdots$

TABLE 3. The four rational weight-5 CM newforms at level $N = 84$, $K = \mathbb{Q}(\sqrt{-21})$, $h(K) = 4$. The invariant $a_5 = 16$ confirms Theorem 3.1(d).

Form	a_2	a_3	a_5	a_7	a_{11}	a_{13}
F1	+4	+9	16	-34	+36	+49
F2	+4	-9	16	+34	-36	-49
F3	-4	+9	16	+34	-36	+49
F4	-4	-9	16	-34	+36	-49

4.3. Class-number-4 field: $D = -84$, $K = \mathbb{Q}(\sqrt{-21})$. The quartet $\{F1, F2, F3, F4\}$ forms a single $\text{Cl}(K)$ -orbit with $\text{Cl}(K) \cong (\mathbb{Z}/2\mathbb{Z})^2$, as required by Theorem 3.1(c). The sign-flip pattern in a_2 and a_3 corresponds to the two independent generators of $\text{Cl}(K)$, one reflecting the class of the prime above 2 and one the class above 3. The coefficient

$a_5 = 16$ is invariant across the orbit because 5 is split in K and the Hecke character value at the prime above 5 is fixed by the CM type.

5. DISTINCTION: RATIONAL EIGENFORM VS. RATIONAL L -VALUE

A central application context for these CM newforms is the study of special L -values. Define, following [4]:

$$\alpha_2(f) := \frac{L(f, 2) \cdot \pi^2}{\Omega_f^4},$$

where Ω_f is the CM period of f normalised so that $\Omega_f^4 \in_{>0}$ and $\alpha_2 \in K^{\text{H}\times}$.

Proposition 5.1 (Goldstein–Schappacher). *With the above notation:*

$$\alpha_2(f) \in \mathbb{Q} \iff h(K) = 1.$$

For $h(K) \geq 2$, $\alpha_2(f)$ lies in $K^{\text{H}} \setminus \mathbb{Q}$ (specifically in the totally real subfield of K^{H}).

Proof. This is [2, Théorème 5.1], combined with the period-quotient interpretation of Shimura–Taniyama. The Galois conjugates $\sigma(\alpha_2)$ for $\sigma \in \text{Gal}(K^{\text{H}}/K)$ produce h distinct values, all algebraically conjugate over $\mathbb{Q}$. They collapse to a single rational value only when $h = 1$. $\square$

Remark 5.2 (NB4 confirmation). The PARI computation for $D = -88$ ($h = 2$) yields:

$$L(f_1, 2) \cdot \pi^2 \approx 62.85\dots, \quad L(f_2, 2) \cdot \pi^2 \approx 24.24\dots$$

at 117 significant digits; neither value is approximable by a rational number with small denominator. This confirms Proposition 5.1 numerically and shows that PARI’s `mfeigenbasis` returning a dimension-1 space (“rational eigenform”) does *not* imply $\alpha_2 \in \mathbb{Q}$. The two notions are entirely distinct: the eigenform is rational (all $a_p \in \mathbb{Q}$), but the L -value quotient is not.

Remark 5.3. For $h(K) = 4$ ($D = -84$), each of the four rational eigenforms F1–F4 has $\alpha_2 \in K^{\text{H}}$ with $[K^{\text{H}} : \mathbb{Q}] = 4$. The four values $\alpha_2(F1), \dots, \alpha_2(F4)$ are a complete set of Galois conjugates over $\mathbb{Q}$.

6. OPEN QUESTIONS: THE $h = 3$ ANOMALY

6.1. Vanishing for $h \equiv 0 \pmod{3}$. The PARI computation via the script `h3_ext.gp` searched for rational CM weight-5 newforms across all eight fundamental discriminants with $h(K) = 3$, namely $D \in$

$\{-23, -31, -59, -83, -107, -139, -211, -283\}$, and found zero rational eigenforms in every case. This is the $h = 3$ *anomaly* referred to in Theorem 3.1(b).

The explanation lies in the structure of $\text{Cl}(K)$ when $3|h(K)$. A 3-element subgroup of $\text{Cl}(K)$ gives rise to cubic character twists, and the associated Grössencharacter values at generic split primes p satisfy $\psi(\mathfrak{p}) \in \mathbb{Q}(\zeta_3) \setminus \mathbb{Q}$. More precisely, if $[\mathfrak{a}]$ has order 3 in $\text{Cl}(K)$, then

$$a_p(f_{\mathfrak{a}}) = \psi_0(\mathfrak{p})\chi_{\mathfrak{a}}(\mathfrak{p}) + \psi_0(\bar{\mathfrak{p}})\chi_{\mathfrak{a}}(\bar{\mathfrak{p}})$$

involves a primitive cube root of unity $\chi_{\mathfrak{a}}(\mathfrak{p}) = \zeta_3$ for primes p that split in K^H/K in a specific way, forcing $a_p \notin \mathbb{Q}$ for such p .

Proposition 6.1. *If $3|h(K)$ then $S_D = \emptyset$.*

Proof. Let $[\mathfrak{c}] \in \text{Cl}(K)$ be an element of order 3. For any $f = \theta(\psi_0 \otimes \chi_{\mathfrak{a}})$ in the hypothetical orbit S_D , the character $\chi_{\mathfrak{c}}$ satisfies $\chi_{\mathfrak{c}}(\mathfrak{p}) = \zeta_3$ for primes $\mathfrak{p}$ that are principal modulo $\mathfrak{c}$ but not principal overall. By Chebotarev’s density theorem, a positive density of primes p split in K with $\chi_{\mathfrak{c}}(\mathfrak{p}) = \zeta_3$ exist, so $a_p(f) \notin \mathbb{Q}$ for those p . Hence $f \notin S_D$. $\square$

6.2. The “cubic-Hecke” forms. Although no member of the $h = 3$ orbit is rational over $\mathbb{Q}$, the LMFDB data shows that the Hecke field of each such form is $\mathbb{Q}(\zeta_3)$ (a degree-2 extension of $\mathbb{Q}$). We call these *cubic-Hecke* forms. Remarkably, the period quotient α_2 for these forms still satisfies a parity-determined formula (companion note in preparation):

$$\frac{\alpha_1(f)}{\sqrt{|D|} \cdot |D|} = \begin{cases} 3/4 & D \equiv 1 \pmod{4}, \\ 6 & D \equiv 0 \pmod{4} \end{cases}$$

verified across all $h = 3$ fields tested, suggesting a deeper invariance of the CM period structure independent of class-number parity.

6.3. Open questions.

- (1) Does Proposition 6.1 extend to all h with $3|h$? The argument given applies to any $[\mathfrak{c}]$ of order 3, but one should check whether higher-order elements of $\text{Cl}(K)$ could conspire to cancel the cubic irrationality.
- (2) Can the $a_5 = 16$ invariance of Theorem 3.1(d) be proved in general for all D such that 5 splits in K with a specific Frobenius structure?
- (3) For $h(K) = 3$ fields, what is the algebraic degree of $\alpha_2(f)$ over $\mathbb{Q}$? Is it always $3 = h(K)$?
- (4) Do analogous orbit-count theorems hold for other weights $k \equiv 1 \pmod{4}$? (Weight 9, infinity type $(8, 0)$, etc.)

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